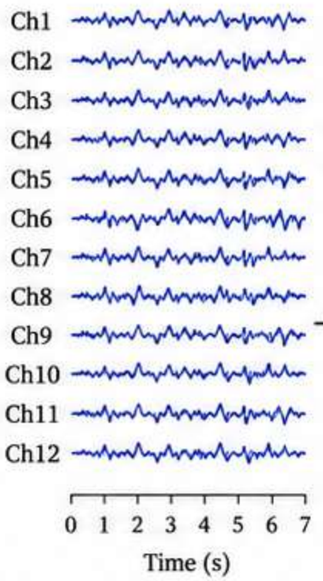
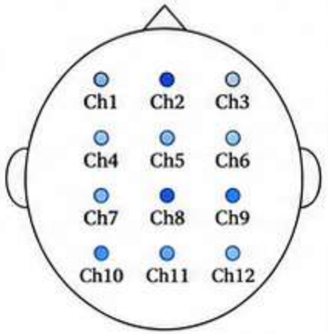
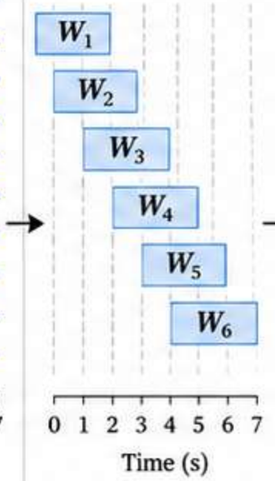
Preprocessed EEG epochs



12 channels, 256 Hz, 7 s epoch

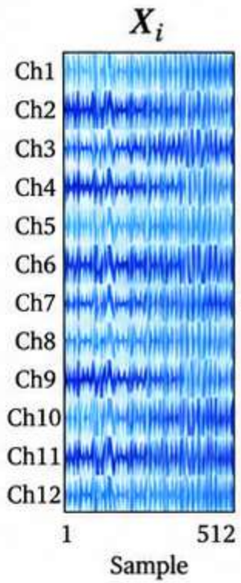


Windowing



2-s windows, 50% overlap, 6 windows/epoch

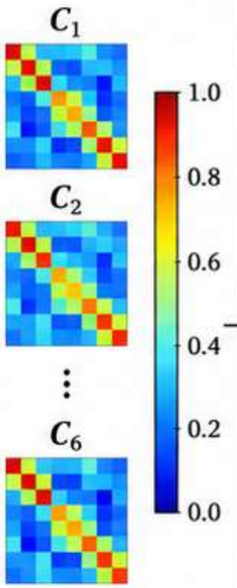
Window-level EEG matrix



$X_i \in \mathbb{R}^{12\times512}$

12 × 512 (12 channels, 2 s × 256 Hz)

Shrinkage covariance estimation

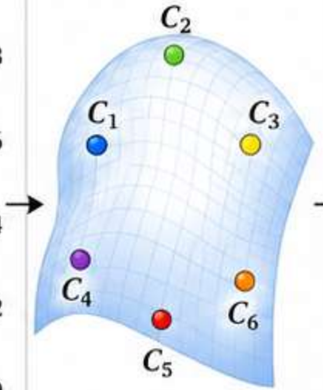


OAS / shrinkage covariance

$C_i \in \mathbb{R}^{12\times12}$

SPD covariance matrices ($C_i > 0$)

Riemannian manifold

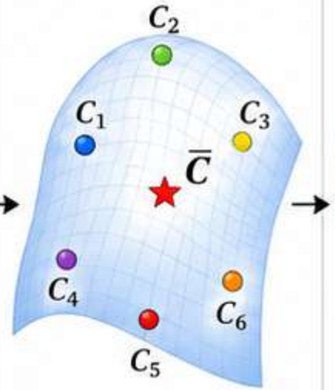


Riemannian manifold of SPD matrices

$C_i \in S_{++}^{12}$

(SPD, 12 × 12)

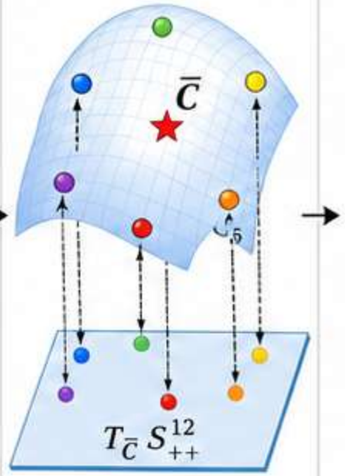
Training reference mean



Riemannian mean (training covariances only)

$$\bar{C} = \arg\min_{C_i \in S_{++}^{12}} \sum_{C_i \in \mathcal{T}} d_R^2(C, C_i)$$

Tangent-space projection



Tangent-space mapping

$T_{\bar{C}}\, S_{++}^{12}$

Feature vector

$$z_i = \begin{bmatrix} z_{i,1} \\ z_{i,2} \\ \vdots \\ z_{i,78} \end{bmatrix}$$

Symmetric vectorization

$z_i \in \mathbb{R}^{78}$

78-dimensional feature vector (12 × 13 / 2)

Classifier

RBF-SVM classification

- Back
- Down
- Forward
- Left
- Right
- Up